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Metal oxide semiconductor with multiple drain vias

Abstract

A metal oxide semiconductor (MOS) with multiple drain vias includes a semiconductor substrate, which is divided into a gate region, a source region and a drain region. On the same surface of the semiconductor substrate in the gate region, a connection layer is formed in the source region and the drain region with a low-resistance metal material. A plurality of conductive elements, such as solder balls, can be directly arranged on the connection layer, so that the MOS can be soldered to the circuit board through the conductive elements distributed on the same surface. In the drain region, there are a plurality of vias. The inside of each via is filled with the connection layer and extends to the inside of the semiconductor substrate. With the aforementioned structure, the MOS has the advantage of low on-resistance $R_{DS(ON)}$.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The present application claims benefit of U.S. provisional application No. 63/350,877, filed on Jun. 10, 2022, under 35 U.S.C. 119(e), the content of which is hereby incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

1. Field of the Invention
- (1) The present application relates to a metal oxide semiconductor (MOS) with multiple drain vias, and particularly to a metal oxide semiconductor element with a horizontal structure.
2. Description of Related Art
- (2) Most metal oxide semiconductor (MOS) chips are formed in a vertical design. The structure of

the MOS chip is shown in FIG. 10. In terms of the contact layout of the chip, a contact of a gate region G and a contact of a source region S are both formed on a top side of the MOS chip, and a drain region D is formed by a metal layer on the backside of the chip. Because the drain region D is located vertically below the gate region G and the source region S, the structure of the MOS chip is called a vertical design.

(3) FIG. 11 is a schematic side view of the mentioned MOS chip. The left region of the figure is the gate region G, and the right region is the source region S. The specific structures of the unit cells in the gate region G and the source region S are omitted from the figure. A metal layer 500 is formed on the surfaces of the gate region G and of the source region S, and the gate region G and the source region S are electrically connected through the metal layer 500.

(4) With reference to FIG. 12, the MOS chip must be packaged such that it can be electrically connected to a circuit board. For example, the backside of the MOS chip (i.e., the drain region D) is attached on a lead frame 510, and the metal layers 500 on the surface of the gate region G and on the surface of the source region S are respectively and electrically connected to different pins of the lead frame 510 through a metal wire 520 by wire bonding. In FIG. 12, the metal layer 500 of the gate region G is also connected to another independent pin through an independent metal circuit, which is omitted from the figure. Finally, the chip is covered with an insulating material 530 to form a complete package of MOS.

(5) MOS chips adopt a vertical design, so the chip must be repackaged before soldering on the circuit board. The drain region D at the bottom of the chip is made of metal and must be electrically connected to a substrate (such as the lead frame 510), and the metal layer 500 with the gate region G located on the top surface of the chip and the metal layer 500 with the source region S located on the top surface of the chip are electrically connected to the corresponding pins respectively through metal wires. Finally, the insulating material 530 is being processed on the chip. Therefore, many processing steps are required to package the MOS.

(6) When the metal layer 500 is used as the conductive metal on the gate region G and the source region S, the on-resistance $R_{DS(ON)}$ of the MOS is difficult to further decrease, and this should be further improved.

SUMMARY OF THE INVENTION

(7) The present invention provides a metal oxide semiconductor (MOS) with multiple drain vias, which can make the contacts of the gate, of the drain and of the source all located on the same surface, and has the characteristics of low on-resistance $R_{DS(ON)}$.

(8) The MOS of the present invention comprises: a semiconductor substrate, having a gate region, a source region and a drain region, wherein, the gate region, the source region, and the drain region each have a respective connection layer made of a metal material and formed on the same surface of the semiconductor substrate, and the connection layers on the gate region, on the source region and the on drain region are insulated and separated from each other; wherein, a plurality of vias extending into the semiconductor substrate are formed in the drain region, and the connection layer is filled in the vias; the connection layers on the gate region, on the source region and on the drain region are each electrically connected to a respective conductive element, and the MOS is electrically connected to the outside through the conductive elements.

(9) The present invention can be fabricated by wafer-level chip scale packaging (WL CSP) technology, and the gate region, the source region and the drain region are made to be horizontally distributed. The MOS is provided with conductive elements on the same surface for external electrical connection. Therefore, there is no need to use a traditional lead frame or operations such as wire bonding and molding.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a three-dimensional schematic diagram of the gate contact, source contact and drain contact layout of the MOS of the present invention;
- (2) FIG. 2 is a surface schematic diagram of the gate contact, source contact and drain contact layout of the MOS of the present invention;
- (3) FIGS. 3A-3G show processing step diagrams of the gate of the MOS of the present invention;
- (4) FIGS. 4A-4F show processing step diagrams of the source of the MOS of the present invention;
- (5) FIGS. 5A-5K show processing step diagrams of the drain of the MOS of the present invention;
- (6) FIG. 6 is a schematic plan diagram of forming an via in the drain region of the present invention;
- (7) FIG. 7 is a three-dimensional schematic diagram of the MOS of the present invention;
- (8) FIG. 8 is a schematic diagram of soldering the MOS to a circuit board in the present invention;
- (9) FIG. 9 is a schematic diagram of soldering a conventional MOS to a circuit board;
- (10) FIG. 10 is a schematic diagram of the contact positions of the gate, source and drain of the conventional MOS;
- (11) FIG. 11 is a cross-sectional diagram of the MOS in FIGS. 10; and
- (12) FIG. 12 is a schematic diagram of the MOS in FIG. 11 after packaging.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENT(S)

- (13) The present invention is a metal oxide semiconductor (MOS) with multiple drain vias, which can be a P-type metal oxide semiconductor (PMOS) or an N-type metal oxide semiconductor (NMOS). In the following description, a power metal oxide semi-field effect transistor (Power MOSFET) is used as an example for elaboration.
- (14) With reference to FIG. 1 and FIG. 2, the MOS of the present invention has a gate region G, a source region S and a drain region D, and these regions are used as contact regions for the MOS to be electrically connected to the outside elements. Different from the traditional vertical layout, the present invention makes the drain region D on the side of the source region S. In this embodiment, the drain region D is located on one side of the source region S, and the gate region G is located at the opposite corner of the source region S, so that the gate region G, the source region S, and the drain region D form a horizontal layout together. As for the area size of each region, the source region S is bigger than the drain region D, and the drain region D is bigger than the gate region G.
- (15) The following describes the cell unit structure in the gate region G, the source region S and the drain region D through the processing. The processes can be roughly divided into gate region fabrication, source region fabrication and drain region fabrication, and these processes are not limited to be performed sequentially. In one embodiment, several fabrication steps of the gate region G and the source region S are performed simultaneously. In the figures, each region is illustrated with 1 to 2 unit cells as an example, and the MOS actually includes a plurality of unit cells.
- (16) The production stage of the gate region:
- (17) With reference to FIG. 3A, a semiconductor substrate **100** is provided. The substrate **100** defines a gate region G, a source region S and a drain region D. The material of the substrate **100** can be selected from silicon, gallium arsenide, silicon germanium, silicon carbon or other semiconductor materials known in the semiconductor field. In this embodiment, an epitaxy layer **102** is grown on the substrate **100**, and an oxide protection layer **104** is covered on the epitaxy layer **102**. In one embodiment, the substrate **100** and the epitaxy layer **102** are the first conductivity type, e.g., N-type doped.
- (18) With reference to FIGS. 3B-3D, in the gate region G and the source region S, the substrate **100** is selectively etched to form grooves **105**. As shown in FIG. 3B, a patterned first mask layer **401** is formed on the oxidation protection layer **104**. The first mask layer **401** defines a gate pattern, and the area not covered by the first mask layer **401** is expected to form the gate. The first mask layer

401 may be a photoresist layer.

(19) As shown in FIG. 3C, the area not covered by the first mask layer **401** is etched to form a groove **105**, and the groove **105** extends into the epitaxy layer but does not penetrate the epitaxy layer **102**. The etching method depends on the material of the first mask layer **401**, such as wet etching, plasma etching, etc.

(20) As shown in FIG. 3D, after the etching is completed, the first mask layer **401** and the oxide protection layer **104** on the substrate **100** are removed. In the gate region G and the source region S, a plurality of grooves **105** have been formed on the epitaxy layer **102**.

(21) As shown in FIG. 3E, a gate insulating layer **106** is formed inside the groove **105** of the epitaxy layer **102**. The gate insulating layer **106** covers the bottom and sides of the groove **105** but does not completely fill the groove **105**. Wherein, the gate insulating layer **106** may be a dielectric material formed by spin coating, heat treatment, or deposition.

(22) As shown in FIG. 3F, a gate conductive layer **107** is deposited on the gate insulating layer **106**. The thickness of the gate conductive layer **107** can fill each groove **105**. Wherein, the material of the gate conductive layer **107** can be selected from polysilicon.

(23) As shown in FIG. 3G, etching back the gate conductive layer **107** such that the gate conductive layer **107** only fills the interior of each groove **105** and is exposed to the epitaxy layer **102**. The gate conductive layer **107** filled in each groove **105** serves as the gate conductive layer **107** of each unit cell.

(24) The production stage of the source region:

(25) After the gate conductive layer **107** is filled in the groove **105**, a source structure is formed in the source region S. In FIGS. 4A to 4F, only the fabrication process of the source region S is shown, and the structural representation of the gate region G and the drain region D is omitted. Referring to FIG. 4A, ion implantation or doping is performed on the epitaxy layer **102** to form a well **201**. The material used for ion implantation or doping has a second conductivity type, such as P-type doping, which is different from the first conductivity type of the epitaxy layer **102**. In one embodiment, the material for ion implantation or doping may be boron.

(26) As shown in FIG. 4B, a patterned second mask layer **402** is formed on the epitaxy layer **102**. The second mask layer **402** is used to define the source pattern, and the area not covered by the second mask layer **402** is expected to form the source. Wherein, the top and both sides of the groove **105** are not covered by the second mask layer **402**. The second mask layer **402** may be a photoresist layer.

(27) As shown in FIG. 4C, using the second mask layer **402** as a mask layer for implantation, ion implantation or doping is performed in the well **201** to form a first implanted region **202** of the source. The material used for ion implantation or doping has a first conductivity type, such as N-type doping, and the first implanted region **202** is adjacent to both sides of the groove **105**. After the ion implantation is completed, an annealing operation is performed to stabilize the implanted material in the first implanted region **202**.

(28) As shown in FIG. 4D, after the annealing is completed, the second mask layer **402** is removed, and an interlayer dielectric **203** is formed on the epitaxy layer **102**. The interlayer dielectric **203** is an insulating material. A patterned third mask layer **403** is then disposed on the interlayer dielectric **203**. The third mask layer **403** exposes the well **201** between two adjacent grooves **105** and part of the first implanted region **202** on both sides thereof.

(29) As shown in FIG. 4E, using the third mask layer **403** as an etching mask, the interlayer dielectric **203** is etched to remove the interlayer dielectric **203** exposed on the third mask layer **403**.

(30) As shown in FIG. 4F, after the etching of the interlayer dielectric **203** is completed, the third mask layer **403** is removed, and heavy ion implantation or heavy doping is performed on the well **201** to form a second implanted region **204**. The material used for ion implantation or doping is of a second conductivity type, e.g. P-type doping. Each second implanted region **204** is located between two adjacent first implanted regions **202**. After the ion implantation is completed, an

annealing step is performed to stabilize the implanted material in the second implanted region **204**.

(31) The production stage of the drain region:

(32) As shown in FIG. 5A, a slot **301** is formed in the drain region D of the substrate **100**. The depth of the slot **301** extends down to the substrate **100**, so the epitaxy layer **102** at the position of the slot **301** is completely removed. The slot **301** can be formed by dry etching, wet etching, laser, tool cutting, etc. The method of forming the slot **301** is not particularly limited.

(33) With reference to FIG. 5B, a first insulating layer **302** is formed on the substrate **100**. The first insulating layer **302** fully covers the gate region G, the source region S and the drain region D, wherein, the thickness of the first insulating layer **302** is sufficient to fill the slot **301** and cover the inner dielectric layer **203** in the gate region G and the source region S. In one embodiment, the material of the first insulating layer **302** may be polyimide (PI).

(34) With reference to FIGS. 5C and 6, in the drain region D, a plurality of vias are formed in the first insulating layer **302**. In this embodiment, the plurality of vias **303** are formed on the entire surface of the drain region D. The vias **303** penetrate through the first insulating layer **302** and extend to the substrate **100**. The shape and arrangement of each via **303** are not limited; for example, each via **303** can be a square hole, a round hole, etc. The arrangement of the vias **303** can be a regular matrix or a staggered arrangement. FIG. 6 only schematically shows that the drain region D has a plurality of circular vias **303** arranged in a staggered manner.

(35) In addition to forming vias **303** in the drain region D, in the gate region G and source region S, vias **303** penetrating the first insulating layer **302** are also formed at positions corresponding to each of the second implanted regions **204**. The method of making the via **303** may include, but is not limited to, laser and etching.

(36) With reference to FIG. 5D, a seed layer **304** is formed on the surface of the first insulating layer **302**, the bottom of each via **303** and the sidewalls of each via **303**, wherein the seed layer **304** is a conductive material (e.g., titanium/copper), which can be fabricated by sputtering.

(37) With reference to FIG. 5E, a fourth mask layer **404** is formed on the seed layer **304**. The pattern of the fourth mask layer **404** is consistent with the distribution of the gate region G, the source region S and the drain region D, and is used to separate the gate region G, the source region S and the drain region D. The fourth mask layer **404** can be a photoresist layer.

(38) With reference to FIG. 5F, a connection layer **305** is covered on the seed layer **304** of the gate region G, the source region S and the drain region D. The material of the connection layer **305** has relatively low resistance compared to aluminum; for example, the connection layer **305** is a copper layer or a copper alloy layer. The connection layer **305** fills the vias **303** in each region and is electrically connected to the seed layer **304**. The connection layers **305** in different regions are separated by the fourth mask layer **404**.

(39) With reference to FIGS. 5G-5H, after the connection layer **305** is fabricated, the fourth mask layer **404** is removed to expose the seed layer **304** originally covered by the fourth mask layer **404**. The connection layers **305** between the gate region G, the source region S and the drain region D are separated from each other. After removing the fourth mask layer **404**, the exposed seed layer **304** is further removed. For example, the seed layer **304** made of titanium/copper is removed by etching, so that the gate region G, the source region S and the drain region D are not directly electrically connected to each other through the seed layer **304**.

(40) With reference to FIGS. 5I-5J, a second protective layer **306** is formed on the surface of the connection layer **305**, and the second protective layer **306** completely covers the gate region G, the source region S and the drain region D. Several contact vias **307** are formed on the second protective layer **306**. Each contact via **307** penetrates through the second protective layer **306** and extends to the connection layer **305**. Wherein, the position and number of the contact vias **307** are determined by the position and the number of the bonding pads required by the MOS. In this embodiment, one contact via **307** is provided in the gate region G, three contact vias **307** are provided in the source region S, and two contact vias **307** are provided in the drain region D.

(41) With reference to FIGS. 5K and 7, each contact via **307** is provided with a conductive element **308**, and the conductive element **308** is electrically connected to the connection layer **305**. In this embodiment, each conductive element **308** is a solder ball, but in other embodiments, the conductive element **308** may also be a flat solder pad. In the gate region G, the conductive element **308** is electrically connected to the gate conductive layer **107** through the connection layer **305**. In the source region S, the conductive element **308** is electrically connected to the second implanted region **204** of the source through the connection layer **305**. In the drain region D, the conductive element **308** is electrically connected to the connection layer **305**, and the connection layer **305** extends to the substrate **100**.

(42) In practical application, the present invention makes the gate region G, the source region S and the drain region D into a horizontal distribution. The MOS forms a contact that can be electrically connected to the outside elements on the same surface. Therefore, it can be fabricated by wafer-level chip scale packaging (WLCSP) technology, without the need for traditional lead frames, wire bonding, molding and other operations, using the conductive element **308** shown in FIG. 8 (such as solder balls) can be directly soldered on the printed circuit board. Compared with the traditional MOS, the volume of the MOS of the present invention can be further reduced.

(43) In the present invention, a plurality of vias **303** (via) extending to the inside of the substrate **100** are formed in the drain region D, and filling the inside of the plurality of vias **303** with a conductive material to form a drain. The on-resistance $R_{sub.DS(ON)}$ of the MOS chip between the source and the drain can be reduced.

(44) Besides, in a preferred embodiment of the present invention, the top metal layer (i.e. the connection layer **305**) of the MOS is made of low-resistance copper-containing metal instead of a traditional metal layer. The conductive elements **308** of the drain region D and of the source region S are connected to the internal circuit of the MOS through the connection layer **305** with low impedance characteristics. When the MOS is turned on by receiving the driving voltage, the on-resistance $R_{sub.DS(ON)}$ between the drain region D and the source region S can be reduced. The signal transmission path between the MOS and the external printed circuit board is also relatively low impedance.

(45) With reference to FIG. 8, it is shown that the drain (D), the source (S), and the gate (G) are fabricated on the same side, and there is no need to rely on the assembly to turn all electrodes to the same side, so the product size can be greatly reduced. Compared with FIG. 9, the dotted line in FIG. 8 indicates that the signal transmission path between the drain (D) and the source (S) can be further reduced, and the conductive media in the transmission path can also be simplified. In the present invention, the conductive element **308** on the chip can be directly connected to the contacts on the circuit board P, so that the overall on-resistance $R_{sub.DS(ON)}$ can be reduced.

(46) The size of the MOS packaged by the packing components of FIG. 9 will be larger than the entire chip, resulting in the lengthening of the conduction path (shown by the dotted line) between the chip and the circuit board P. Moreover, the conductive media in the transmission path may pass through multiple layers (such as solder **540**, metal wires **520**, etc.), so the overall on-resistance $R_{sub.DS(ON)}$ will be high.

(47) Although the present invention has been illustrated and described herein with reference to preferred embodiments and specific examples thereof, it will be readily apparent to those of ordinary skill in the art. All such equivalent embodiments and examples are within the spirit and scope of the present invention, are contemplated thereby, and are intended to be covered by the following claims.

Claims

1. A metal oxide semiconductor (MOS) with multiple drain vias, comprising: a semiconductor substrate, having a gate region, a source region and a drain region, wherein, the gate region, the

source region, and the drain region each has a respective connection layer made of a metal material and formed on a same surface of the semiconductor substrate; the semiconductor substrate having an epitaxy layer with a slot formed through the epitaxy layer, wherein a first protective layer fills the slot; a plurality of vias extending into the semiconductor substrate in the drain region, extending through the first protective layer and the epitaxy layer, and being filled with the connection layer; a second protective layer formed on the connection layers of the gate region, the source region and the drain region; and a plurality of contact vias formed through the second protective layer; wherein, the connection layers being coplanar to each other on the gate region, on the source region and on the drain region are insulated and separated from each other by the second protective layer, and are each electrically connected to a respective conductive element in the contact via; and the MOS is electrically connected to the outside through the conductive elements on the same surface.

2. The MOS with multiple drain vias as claimed in claim 1, wherein the connection layer is a copper layer or a copper alloy layer.

3. The MOS with multiple drain vias as claimed in claim 1, wherein the first protective layer extends to cover the epitaxy layer of the drain region and the source region.

4. The MOS with multiple drain vias as claimed in claim 1, wherein in the gate region, a plurality of gate conductive layers are formed in the epitaxy layer, and each of the gate conductive layers is electrically connected to the connection layer on the gate region.

5. The MOS with multiple drain vias as claimed in claim 4, wherein in the source region, a plurality of unit cells are formed in the epitaxy layer, and each unit cell includes a gate conductive layer and a first implanted region on both sides thereof; and a second implanted region is disposed between two adjacent said unit cells, and the connection layer in the source region electrically contacts the first implanted region and the second implanted region.

6. The MOS with multiple drain vias as claimed in claim 1, wherein each of the conductive elements is a solder ball or a flat pad.
